

CS528 / DSA428 Big Data Processing

Storing Big Data : Apache Hadoop

- Introduction to Apache Hadoop and the Hadoop Ecosystem
- Apache Hadoop Overview
- Apache Hadoop Cluster Components
- HDFS Architecture
- YARN Architecture

Processing Big Data: Apache Spark

- What is Apache Spark?
- Getting Started with RDD and DataFrames
- RDD and DataFrame Operations
- Spark SQL

Introduction to Machine Learning

- Supervised / Unsupervised Learning
- Training ML models
- Parameter Tuning, Cross Validation
- Machine Learning Pipelines

Machine Learning on Big Data : Apache Spark ML

- Spark MLlib Introduction
- Regression, Classification and Clustering with Apache Spark
- Feature Selection and Vectorizing
- Parameter Tuning, Cross Validation

Processing Data Streams : Apache Spark Streaming & Kafka

- Apache Spark Streaming Overview
- Sliding Window Operations
- Structured Streaming
- Apache Kafka Data Sources
- Kafka Topics
- Kafka Producers and Consumers

Grading Policy

- Homework-1 ----- %20
- Homework-2 ----- %20
- Midterm Exam ----- %30
- Final Exam ----- %30